NUMBER-THEORETIC PRIMITIVES: DISCRETE LOGS AND FRIENDS

Building Asymmetric Schemes

- As no encryption scheme besides the OneTimePad is unconditionally secure, we need to find some building blocks — hard problems (assumptions) to base security of our new encryption schemes on.
- Blockciphers and their PRF security is not an option since now we don’t have shared keys in the public-key (asymmetric-key) setting.
- Let’s consider the discrete log related problems and (later) the RSA problem.

Discrete-Log Related Problems

- Let G be a cyclic group and let $m = |G|$. The discrete logarithm function $DLog_{G,g}(a) : G \to \mathbb{Z}_m$ takes $a \in G$ and returns $i \in \mathbb{Z}_m$ such that $g^i = a$.
- There are several computational problems related to this function:
  - Discrete-logarithm (DL) problem
  - Computational Diffie-Hellman (CDH) problem
  - Decisional Diffie-Hellman (DDH) problem

Protocol with AE	Given	Figure out
Discrete logarithm (DL)	g^x	x
Computational Diffie-Hellman (CDH)	g^x, g^y	g^{xy}
Decisional Diffie-Hellman (DDH)	g^x, g^y, g^z	Is $z \equiv xy (\mathrm{mod}\ \|G\|)$?

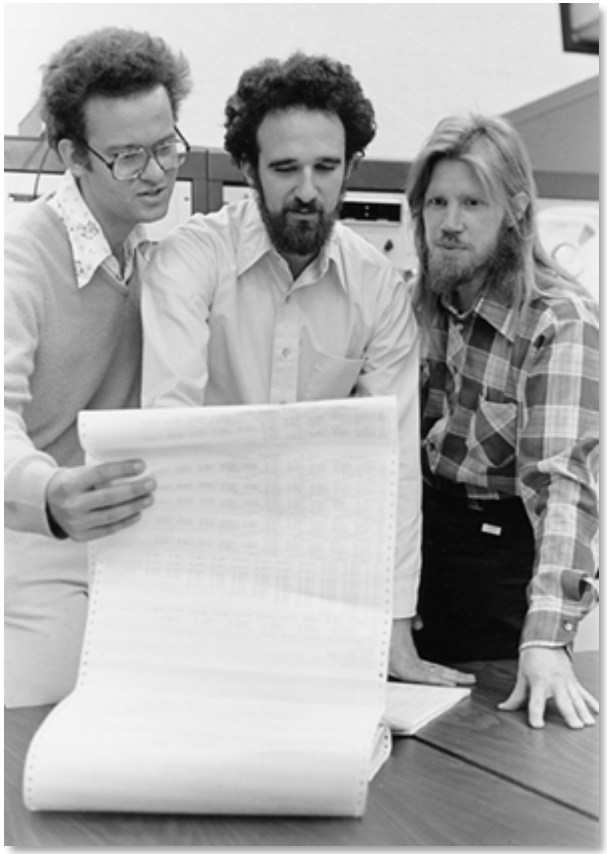
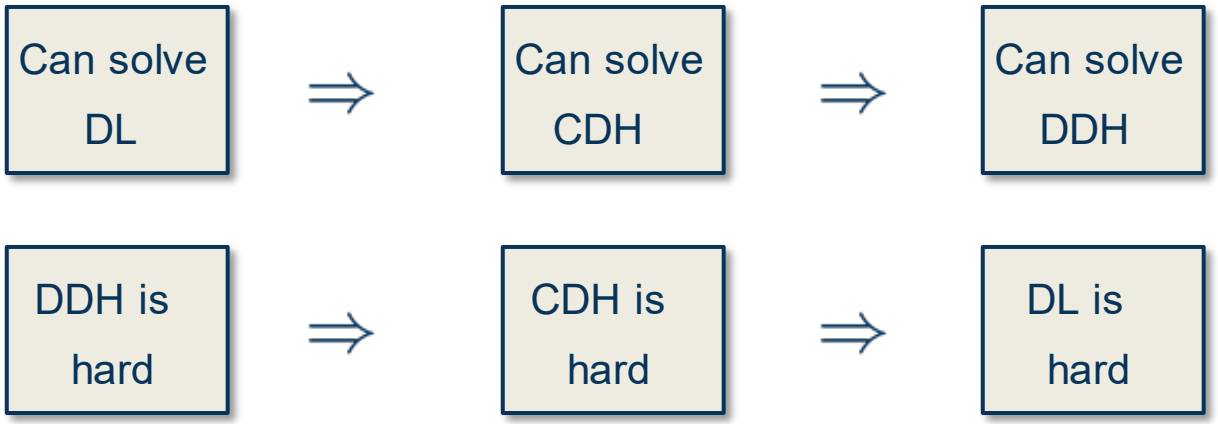


Image courtesy of Stanford University

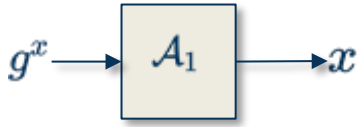
Relations between Problems

- Fix a group and a generator

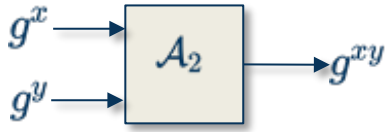


DL Implies CDH

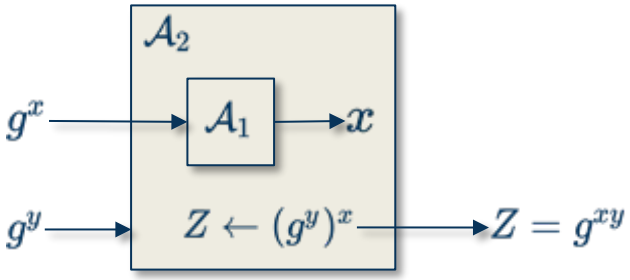
Given : DL solver $\mathcal{A}_1$



Want : CDH solver $\mathcal{A}_2$

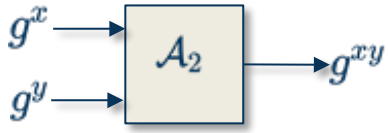


Construction :

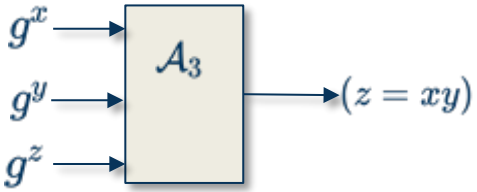


CDH Implies DDH

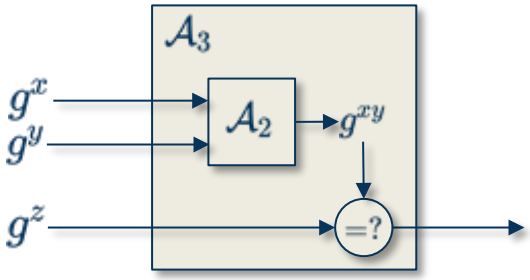
Given : CDH solver $\mathcal{A}_2$



Want : DDH solver $\mathcal{A}_3$



Construction :



DL Problem

- **Def:** Let G be a cyclic group and let $m = |G|$. Let g be a generator. Consider the following experiment associated with an adversary $\mathcal{A}$

$$\begin{array}{l}\textbf{Experiment } \mathbf{Exp}^{\mathsf{dl}}_{G,g}(\mathcal{A}) \\ \quad x \xleftarrow{\$} \mathbb{Z}_m; X \leftarrow g^x \\ \quad \bar{x} \leftarrow \mathcal{A}(X) \\ \quad \text{If } g^{\bar{x}} = X \text{ then return 1 else return 0}\end{array}$$

- The dl-advantage of $\mathcal{A}$ is defined as

$$\mathbf{Adv}^{\mathsf{dl}}_{G,g}(\mathcal{A}) = \Pr[\mathbf{Exp}^{\mathsf{dl}}_{G,g}(\mathcal{A}) = 1]$$

- The discrete logarithm problem is said to be hard in G if the dl-advantage of any adversary with polynomial resources is negligible.

CDH Problem

- **Def:** Let G be a cyclic group of order m. Let g be a generator. Consider the following experiment associated with an adversary $\mathcal{A}$.

$$\begin{array}{l}\text{Experiment } \mathbf{Exp}^{\text{cdh}}_{G,g}(\mathcal{A}) \\ \quad x \xleftarrow{\$} \mathbb{Z}_m; y \xleftarrow{\$} \mathbb{Z}_m \\ \quad X \leftarrow g^x; Y \leftarrow g^y \\ \quad Z \leftarrow \mathcal{A}(X, Y) \\ \quad \text{If } Z = g^{xy} \text{ then return 1 else return 0}\end{array}$$

- The cdh-advantage of $\mathcal{A}$ is defined as

$$\mathbf{Adv}^{\text{cdh}}_{G,g}(\mathcal{A}) = \Pr[\mathbf{Exp}^{\text{cdh}}_{G,g}(\mathcal{A}) = 1]$$

- The computational Diffie-Hellman (CDH) problem is said to be hard in G if the cdh-advantage of any adversary with polynomial resources is negligible.

DDH Problem

- **Def:** Let G be a cyclic group of order m. Let g be a generator. Consider the following experiment associated with an adversary $\mathcal{A}$.

$$
\begin{array}{l|l}
\text{Experiment } \mathbf{Exp}^{\text{ddh-1}}_{G,g}(\mathcal{A}) & \text{Experiment } \mathbf{Exp}^{\text{ddh-0}}_{G,g}(\mathcal{A}) \\
\quad x \xleftarrow{\$} \mathbb{Z}_m & \quad x \xleftarrow{\$} \mathbb{Z}_m \\
\quad y \xleftarrow{\$} \mathbb{Z}_m & \quad y \xleftarrow{\$} \mathbb{Z}_m \\
\quad z \leftarrow xy \mod m & \quad z \xleftarrow{\$} \mathbb{Z}_m \\
\quad X \leftarrow g^x; Y \leftarrow g^y; Z \leftarrow g^z & \quad X \leftarrow g^x; Y \leftarrow g^y; Z \leftarrow g^z \\
\quad d \leftarrow \mathcal{A}(X, Y, Z) & \quad d \leftarrow \mathcal{A}(X, Y, Z) \\
\quad \text{Return } d & \quad \text{Return } d
\end{array}
$$

- The ddh-advantage of $\mathcal{A}$ is defined as

$$\mathbf{Adv}^{\text{ddh}}_{G,g}(\mathcal{A}) = \Pr[\mathbf{Exp}^{\text{ddh-1}}_{G,g}(\mathcal{A}) = 1] - \Pr[\mathbf{Exp}^{\text{ddh-0}}_{G,g}(\mathcal{A}) = 1]$$

- The decisional Diffie-Hellman (DDH) problem is said to be hard in G if the ddh-advantage of any adversary with polynomial resources is negligible.

Hardness of the Problems

- The computational complexities of the problems depend on the choice of a group:

Problem	Group	
	$\mathbb{Z}_p^*$	EC
DL	hard	harder
CDH	hard	harder
DDH	easy	harder

hard: best known algorithm takes time $e^{1.92(\ln p)^{1/3}(\ln \ln p)^{2/3}}$

harder: best known algorithm takes time $\sqrt{p}$, where p is the prime order of the group.

easy: There is a polynomial time algorithm

- If the prime factorization of order of the group is known, e.g., for $\mathbb{Z}_p^*$, if $p-1 = p_1^{\alpha_1} \dots p_n^{\alpha_n}$

  then the DL problem can be solved in time in the order of

$$\sum_{i=1}^{n} \alpha_i \cdot (\sqrt{p_i} + |p|)$$

- Thus if we want the DL problem to be hard, then at least one prime factor needs to be large. E.g. $p = 2q + 1$ where q is a large prime.

DDH is Easy in $\mathbb{Z}_p^*$

- Why?
  - Squares vs non-squares…

- **Claim:** [DDH is easy in $\mathbb{Z}_p^*$]. Let $p \geq 3$ be a prime, let $G = \mathbb{Z}_p^*$, and let g be a generator of G. Then there is an efficient adversary $\mathcal{A}$ with DDH advantage 0.5.

- **Proof:** The idea is to compute and analyze the Legendre symbols of the inputs.

$$\begin{array}{l}\text{Adversary } \mathcal{A}(X,Y,Z) \\ \quad \text{If } J_p(X)=1 \text{ or } J_p(Y)=1 \\ \quad\quad \text{Then } s \leftarrow 1 \text{ Else } s \leftarrow -1 \\ \quad \text{If } J_p(Z)=s \text{ then return 1 else return 0}\end{array}$$

- We claim that

$$\Pr[\mathbf{Exp}^{\text{ddh-1}}_{G,g}(\mathcal{A})=1]=1$$

$$\Pr[\mathbf{Exp}^{\text{ddh-0}}_{G,g}(\mathcal{A})=1]=\tfrac{1}{2} \quad .$$

Subtracting and noting that computing the Legendre symbol takes cubic time in $|p|$ (computed via exponentiation) we get the statement.

- We often want the DDH problem to be hard
- The DDH problem is believed to be hard in several groups, e.g.
  - QR($\mathbb{Z}_p^*$) - the subgroup of quadratic residues of $\mathbb{Z}_p^*$ where $p = 2q + 1, \quad p, q$ are primes. (It's a cyclic group of prime order)
  - Elliptic curves